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Significantly enhanced piezoelectricity in low-temperature sintered Aurivillius-type ceramics with ultrahigh Curie temperature of 800 °C

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Abstract

We report an Aurivillius-type piezoelectric ceramic (Ca_{1-2x} $(LiCe)_xBi_4Ti_{3.99}Zn_{0.01}O_{15}$) that has an ultrahigh Curie temperature (T_c) around 800 °C and a significantly enhanced piezoelectric coefficient (d_{33}), comparable to that of textured ceramics fabricated using the complicated templating method. Surprisingly, the highest d_{33} of 26 pC/N was achieved at an unexpectedly low sintering temperature (T_s) of only 920 °C (~200 °C lower than usual) despite the non-ideal density. Study of different synthesized samples indicates that a relatively low T_s is crucial for suppressing Bi evaporation and abnormal grain growth, which are indispensable for high resistivity and effective poling due to decreased carrier density and restricted anisotropic conduction. Because the layered structure is sensitive to lattice defects, controlled Bi loss is considered to be crucial for maintaining structural order and spontaneous polarization. This low-T_s system is very promising for practical applications due to its high piezoelectricity, low cost and high reproducibility. Contrary to our usual understanding, the results reveal that a delicate balance of density, Bi loss and grain morphology achieved by adjusting the sintering temperature is crucial for the enhancing performance in Aurivillius-type high-T_c ceramics.

Keywords: ferroelectrics, piezoelectricity, Curie temperature, hysteresis loop, doping

(Some figures may appear in colour only in the online journal)

1. Introduction

High-temperature sensing based on the electromechanical response of piezoelectric materials is of significant importance for the chemical engineering, automotive and aerospace industries [1, 2]. The ever-increasing demands on operational temperature have spurred a tremendous research effort on high-temperature piezoceramics. When an operational temperature of 400 °C or higher is required, bismuth layer-structured ferroelectrics (BLSFs) or Aurivillius-type piezoceramics seem to be the only available materials because of their ultrahigh Curie temperature (T_c) of around 600–900 °C, in addition to their low cost and non-toxicity [1, 2]. BLSFs consist of regular perovskite-like $(A_{m-1}B_mO_{3m+1})^{2-}$ slabs separated by $(Bi_2O_2)^{2+}$ layers, where m is the number of perovskite-like slabs. BLSFs also have low temperature coefficients of resonance frequency, low aging rates, and excellent fatigue endurance, etc. However, practical applications of BLSFs are severely limited due to their very low piezoelectricity and difficulty of dense ceramic fabrication, because both spontaneous polarization (along the a-b plane) and mass transport during sintering are restricted by their layered structure [3, 4]. To resolve these problems, several strategies have been tried, including composition adjustment (doping) [5, 6], intergrowth or fabrication of compounds [7–9], and special techniques like hot pressing or fabrication of textured ceramics [10, 11]. The last two strategies usually have associated problems such as anisotropic performance, poor mechanical properties and high cost, etc [4, 7, 10]. In comparison, doping is more feasible for practical industrial fabrication as it may largely modify the sintering activity, microstructure and electrical properties of the ceramics without obvious side effects.

A common concern when using BLSFs is the volatilization of Bi during processing (Bi_2O_3 has a T_m of 820 °C), which generates oxygen vacancies [12]: $Bi_2O_3 = 2Bi + 3/2O_2 + 2V_{Bi}''' + 3V_O^{\cdot\cdot}$, and hence inducesa *p*-type conduction mechanism in pure BLSF materials [13, 14]. Because of the high coercive field of BLSFs, high resistivity is essential both for high-temperature applications and for effective poling, which is necessary to achieve piezoelectricity. Therefore, control of Bi content is crucial for BLSF fabrication. An excess amount of Bi_2O_3 has been used to compensate for Bi evaporation in BLSF ceramic fabrication [15]. However, this leads to difficulty in controlling the stoichiometry of the composition and in turn to poor reproducibility. Low-temperature sintering has been used for various electronic ceramics [16] owing to its many advantages, such as low cost, ease of processing, low environmental pollution, and compatibility with multilayer structures. However, its application to BLSFs seems to be very rarely reported. In contrast, a high sintering temperature is generally considered to be necessary due to the difficulty of dense BLSF ceramic fabrication [3]. $CaBi_4Ti_4O_{15}$ (CBT) is a BLSF-type material with a high T_c of about 790 °C. However, compared to other $m = 4$ BLSF ceramics, e.g. $K_{0.5}Bi_{4.5}Ti_4O_{15}$ ($T_c = 555$ °C, $d_{33} = 21.2$ pC/N), CBT has a low d_{33} of only about 7–8 pC/N [17]. Here we report low-temperature sintered codoped $Ca_{1-2x}(LiCe)_xBi_4Ti_{3.99}Zn_{0.01}O_{15}$ ceramics using $LiCO_3$ as a raw additive that can also act as a sintering aid. Importantly, due to its well-controlled composition and high resistivity, a significantly increased d_{33} of 26 pC/N (about four times of that of pure CBT) was obtained for an optimum composition sintered at a temperature of only 920 °C, which is 200 °C lower than those in existing reports [5, 6, 11]. This study reveals that a delicate balance of density, Bi loss and grain morphology at a relatively low sintering temperature is crucial for enhancing the performance of Aurivillius-type high-T_c ceramics. It may also provide some new hints on how to enhance their performance.

2. Experimental procedure

The study is focused on samples with a Li–Ce codoping mole ratio of 0.025, which exhibit the highest d_{33} according to experiments on different $Ca_{1-2x}(LiCe)_xBi_4Ti_{3.99}Zn_{0.01}O_{15}$ (hereafter referred to as CBT-*x*LiCe, $x = 0$–0.075) ceramics. For sample preparation, a stoichiometric ratio of analytically pure $CaCO_3$, Bi_2O_3, Li_2CO_3, CeO_2, and commercial grade TiO_2 and ZnO powders were ball-milled in ethanol for 1000 min. Then the slurry was dried, ground, and calcined

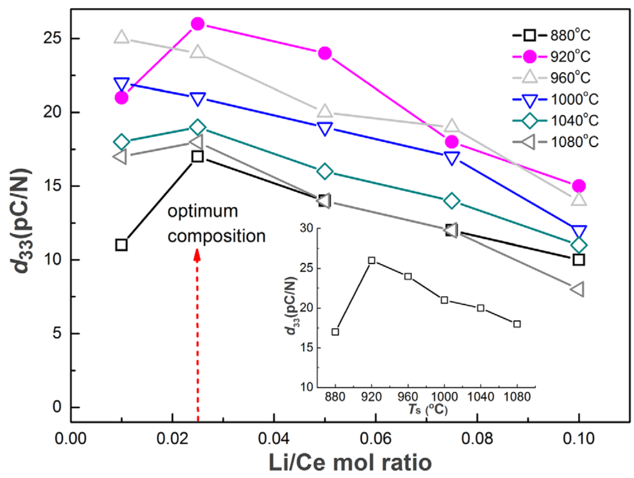

Figure 1. Doping level dependence of d_{33} values of various CBT-*x*LiCe samples sintered at different temperatures.

Table 1. Characteristic electrical parameters of the CBT-0.025LiCe samples.

T_s	880	920	960	1000	1040	1080
ε_r at 1 kHz	62.36	91.43	228.17	132.01	156.58	172.63
tan δ	0.036	0.028	0.032	0.022	0.028	0.022
Q_m	752	953	926	987	921	814
K_p	0.53	0.062	0.059	0.063	0.059	0.058

at 800 °C for 150 min. The final powders were mixed with PVA and pressed into pellets with a diameter of ~10 mm and a thickness of ~1 mm, and then sintered at temperatures from 880 °C to 1080 °C. Densities were measured using the Archimedes method. X-ray diffraction (XRD) profiles were characterized by a Bruker D8 Advance diffractometor with Cu Kα1 radiation. The morphology of the fresh fractured surfaces of the samples was examined by scanning electron microscopy (SEM, Hitachi S-4800, Japan). Ceramic compositions were also derived by comparing the x-ray fluorescence (XRF) peak intensity of the elements using a Skyray EDX3600B instrument. The temperature dependence of the dielectric constant ε_r (1 kHz–1 MHz) of unpoled samples with silver electrodes was measured using a HP4294A impedance analyzer. After poling at 180 °C for 20 min under an electric field of about 9 kV mm^{-1} in silicone oil, d_{33} values were tested using a ZJ-3A type d_{33} meter (Institute of Acoustics, Chinese Academy of Sciences). The mechanical quality factor (Q_m) and planar electromechanical coupling factor (K_p) were measured via the resonance–anti-resonance method using the impedance analyzer. *In situ* high-temperature k_p values were derived by the same method at each temperature. The polarization hysteresis loops were measured using a Radiant Precision LC ferroelectric testing system. A HP4140B pA Meter with a HP16055A Test Fixture was used to measure resistivity. Thermal depolarization experiments were also conducted by annealing poled samples at various temperatures for 1 h and then measuring the room temperature d_{33}.

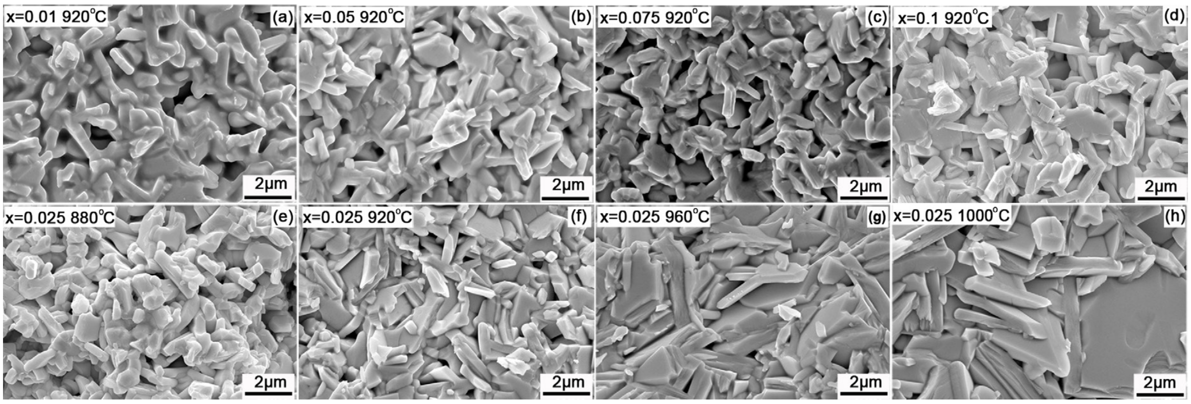

Figure 2. SEM images of typical CBT-*x*LiCe samples; (a)–(d) and (f) show the morphology of samples with different LiCe codoping level sintered at 920 °C; (e)–(h) show the influence of sintering temperatures on the morphology of the optimum CBT-0.025LiCe sample.

3. Results and discussion

The d_{33} values of the CBT-*x*LiCe samples that were sintered at different temperatures are shown in figure 1. The samples show d_{33} in the range of 8 to 26 pC/N. A very interesting phenomenon is that, for all compositions, the maximum d_{33} was achieved at only 920 °C–960 °C, while the highest d_{33} was obtained in the CBT-0.025LiCe sample that was sintered at 920 °C, which is about 200 °C lower than the usually reported data [5, 6, 11]. This is surprising because a high T_s is generally accepted to be indispensable for high density and strong piezoelectricity, while such a low T_s is very desirable for industrial applications due to the associated much lower cost and higher reproducibility. The highest d_{33} of CBT-0.025LiCe (26 pC/N) is about four times of that of pure CBT and comparable to textured CBT that is fabricated with a much more complicated procedure [11]. Therefore, we focus on this composition. Other electrical parameters of the optimum samples, including k_p, room temperature ε_r, etc., are listed in table 1.

Typical SEM images of fresh fractured surfaces of different CBT-*x*LiCe samples are shown in figure 2. For the 920 °C sintered samples, with increasing dopant content the grain morphology changes gradually from a state without a clear boundary to a more densely packed one with polyhedral shapes. The curved grain morphology of the CBT-0.01LiCe sample in figure 2(a) may indicate the presence of a liquid phase due to a low melting point substance, which might disappear at the final sintering state for samples with a higher Li–Ce codoping level. The transitory liquid phase may be due to both Bi_2O_3 and $LiCO_3$, which are commonly used sintering aids. Defects such as oxygen vacancies introduced by the dopants may also accelerate the kinetics. Therefore, the dopants have an obvious effect on facilitating sintering and densification. For the optimum CBT-0.025LiCe composition, the grain size of the differently sintered samples increases gradually with increasing T_s, indicating that a higher temperature may promote grain growth. This might partly explain the decreased piezoelectricity at high T_s because BLSF ceramics with larger grain size have been found to have smaller values of d_{33} [3]. The 920 °C sintered CBT-0.025LiCe sample consists of densely packed grains, with polyhedral geometry, with a size of several microns. However, when T_s is higher, much larger lamellar grains appear, indicating that abnormal grain growth and texture start to occur. These results indicate that the optimum T_s should be around 920–960 °C.

Figure 3(a) shows the XRD patterns of the CBT-0.025LiCe samples that were sintered at different temperatures. The diffraction peaks closely correlate to a $CaBi_4Ti_4O_{15}$ (PDF#52-1640) main phase with a typical orthorhombic symmetry without obvious dependence on T_s. This indicates that the modified CBT-0.025LiCe ceramics have a wide range of sintering temperatures, which is very desirable in terms of reproducibility. The lattice parameters of the samples derived from the XRD patterns and the calculated unit cell volume are listed in table 2. It can be seen that the peaks shift towards higher 2θ angles up to a sintering temperature of 960 °C and towards lower 2θ angles at higher temperatures. This is consistent with the data in table 2, although the reason is unclear.

The relative density, based on the theoretical density calculated from the cell volume, and the DC resistivity of different CBT-0.025LiCe samples are shown in figure 3(b). With increasing T_s, the relative density first increases considerably and then reaches a relatively stable value of around 0.93 above 960 °C, comparable to values reported in the literature [6, 10]. These are consistent with the morphology features shown by the SEM images in figures 2(e)–(h). The resistivity shows a similar dependence on T_s, with the difference that it reaches a plateau at a slightly lower T_s of 920 °C and decreases slightly after reaching a maximum of around 1.38×10^{12} Ω·cm at 960 °C despite the increase in density. The slight decrease in resistivity above 960 °C might be attributed to the enhanced evaporation of Bi_2O_3, which is reported to be volatile above 900 °C [12]. The Bi loss results in more oxygen vacancies and hence may increase the *p*-type conduction. The increase in aspect ratio *L*/*T* of the plate-like grains may also enhance charge transport due to the enhanced contribution from the *a*-*b* plane (which has higher conductivity and is perpendicular to the plate grain). Furthermore, by comparing figure 3(b) and the inset of figure 1, we see that the piezoelectricity shows a similar trend with T_s, indicating a certain degree of correlation (note that the resistivity is on a logarithmic plot). This is partly due to the more effective poling for a highly resistive sample. On the other hand, a sample with small pores may have higher piezoelectricity because it has more grain–pore

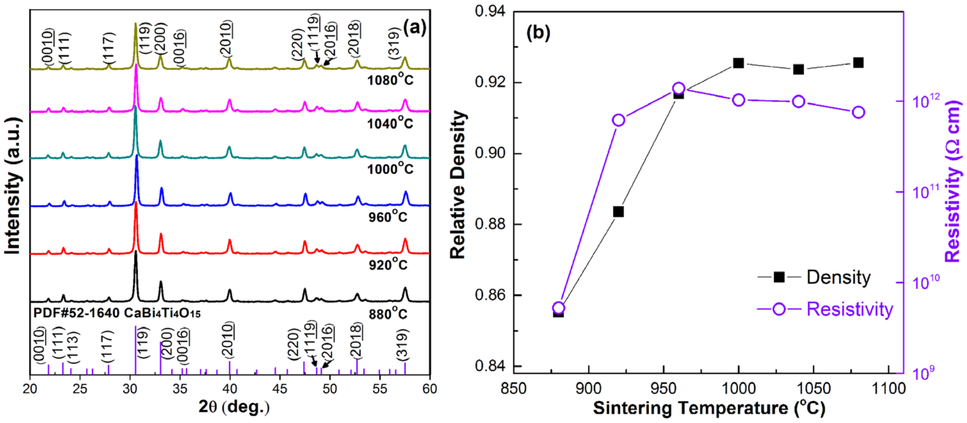

Figure 3. (a) XRD patterns of CB-0.025LiCe samples sintered at different temperatures, where the reference JCPDS card PDF#52-1640 for CBT is also shown at the bottom of the figure; (b) relative density and DC resistivity of CBT-0.025LiCe samples sintered at different temperatures.

Table 2. Sintering temperature dependence of the lattice parameters of the CBT-0.025LiCe samples.

T_s	880	920	960	1000	1040	1080
a (Å)	5.412	5.414	5.406	5.420	5.415	5.426
b (Å)	5.416	5.415	5.411	5.420	5.414	5.418
c (Å)	40.707	40.688	40.576	40.746	40.666	40.674
Cell vol. (Å^3)	1193.21	1192.82	1186.92	1196.95	1192.32	1195.55

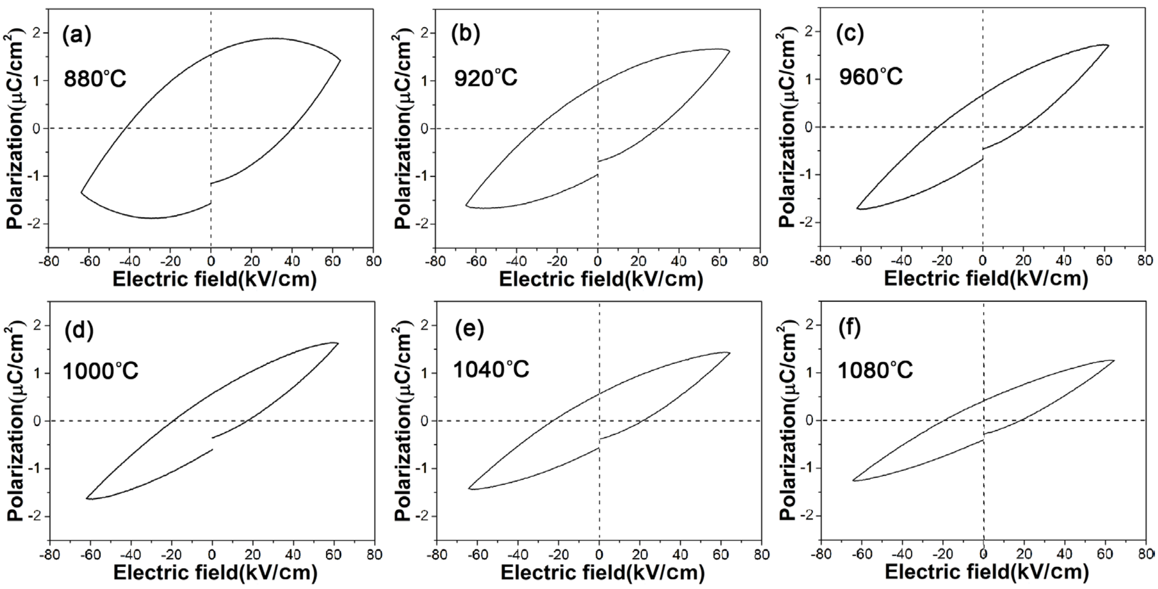

Figure 4. Hysteresis loops of the CBT-0.025LiCe samples sintered at different temperatures.

contact area and thus more of a chance to relax the stress caused by the domain wall motions [18]. These results show that the optimum sintering temperature of 920 °C (with the highest d_{33}) is achieved by the balance of Bi evaporation, densification and grain morphology (conduction anisotropy).

The polarization–electric field (P–E) hysteresis loops of the different CBT-0.025LiCe samples are shown in figure 4. Under the applied voltage all the loops are far from saturation, showing similar elliptical shapes that may indicate similar polarization switching behaviors. The tilted narrow loops imply that the movement of the domains is greatly affected by the external electronic field as is the case with other types of high-T_c ceramics [19]. The round loop of the 880 °C sintered sample indicates a relatively large leakage current that may be due to its relatively loose structure (see figure 2(a)). Except for this sample, both remnant polarization (P_r) and coercive field (E_c) data decrease with increasing T_s after reaching maximum values at 920 °C ($P_r = 0.921$ μC cm^{-2} and $E_c = 29.68$ kV cm^{-1}). According to the electrostrictive coupling relationship [20] $d_{33} = 2\varepsilon_{33}Q_{33}P_r$, where Q_{33} is the electrostrictive coefficient, and ε_{33} is the dielectric coefficient. This explains why the T_s dependence of the P_r data agrees well with the piezoelectric response shown in figure 1. It also reveals that the change in piezoelectricity in differently sintered samples

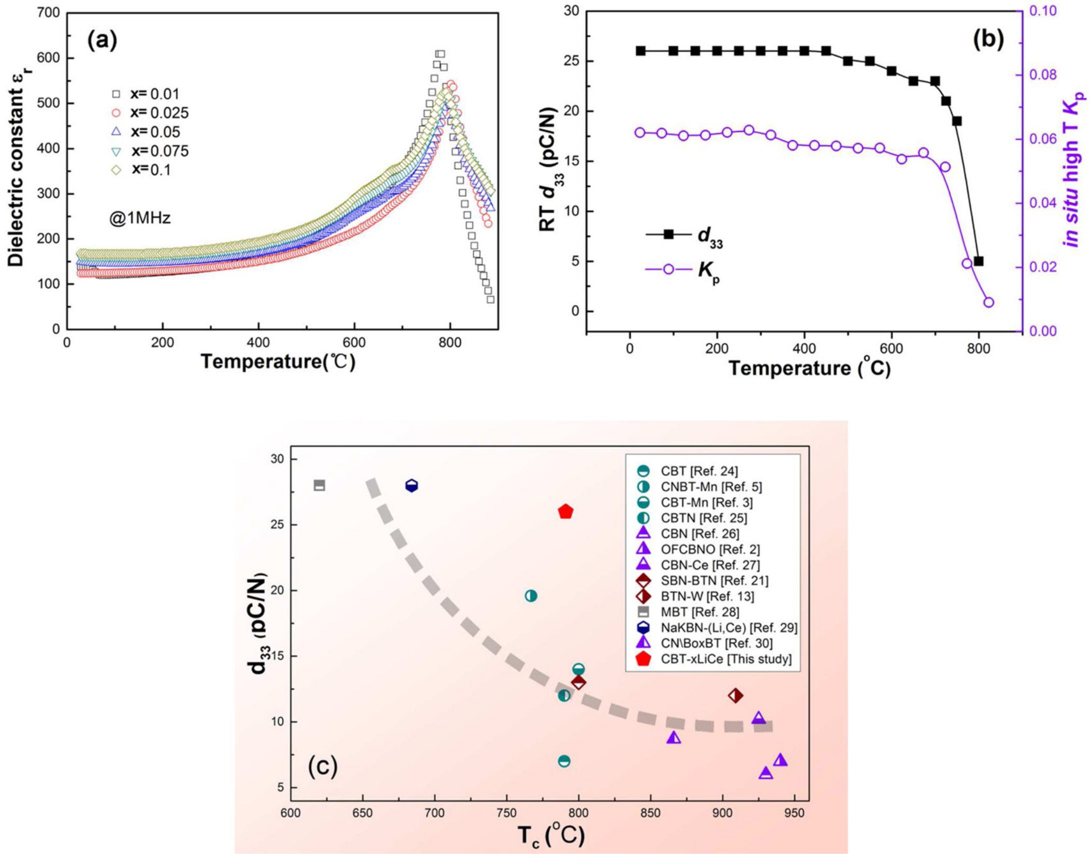

Figure 5. (a) Temperature dependence of dielectric constant (1 MHz) of the different CBT-*x*LiCe samples; (b) variation of d_{33} of the optimum CBT-0.025LiCe samples after annealing (left *y*-axis) and *in situ* high-temperature K_p data (right *y*-axis), (c) a summary of literature data about the correlation of T_c and d_{33} of various Aurivillius ceramics.

mainly come from the variation in the intrinsic polarization contribution, although extrinsic domain wall motion may also contribute to the piezoelectric response.

The T_c of a ferroelectric ceramic is determined by its lattice structure and symmetry; therefore, it may be affected by its material composition but not by the sintering temperature. The T_c values as reflected by the temperature dependence of the relative dielectric constant (1 MHz) of the different CBT-*x*LiCe samples are shown in figure 5(a). The curves of all samples show a sharp peak around 781 °C to 801 °C (the T_c of pure CBT is 790 °C), indicating that the codoped system has a relatively high T_c that shows almost no dependence on the dopant in the tested range. A weak anomaly can also be found around 690 °C, similar to previous observations [21], which may be due to oxygen vacancies and the associated defect dipoles at the grain boundaries or domain walls, which are considered to be constituted as $V_O'' - V_{Bi}''' -$ or $V_O'' - O_{ad}''$ (O_{ad}'' represents absorbed oxygen ions). They are frozen at room temperature and activated at elevated temperatures, contributing to an increase in dielectric permittivity.

The high-temperature stability of the piezoelectricity of the optimum CBT-*x*LiCe sample can be evaluated by its d_{33} data after thermal depolarization and *in situ* high-temperature K_p data as shown in figure 5(b). Usually, the d_{33} of Aurivillius-type piezoceramics continuously decreases after annealing and depolarization may start at a temperature much lower than T_c [5, 6, 22]. While figure 5(b) indicates that the d_{33} of the present CBT-LiCe system is relatively stable, a sharp drop in d_{33} occurs at a temperature very close to the T_c obtained by dielectric measurement, indicating that the depolarization in the system does not occur until very close to Curie point. Judging from the ionic radius (r_i), Li^{+1} ($r_i = 90$ pm) and Ce^{4+} ($r_i = 101$ pm) can replace the A-site Ca^{2+} ($r_i = 114$ pm) or Bi^{3+} ($r_i = 117$ pm), leading to a stronger Coulomb force than with Ca^{2+} or Bi^{3+} with the valance electron of the apex O^{2-} because of their smaller r_i [23]. This may affect the distortion of the perovskite layer octahedra and modify the dipole order, leading to enhanced thermal stability. To make a schematic comparison of the performance of the ceramics of this study with other materials, we summarize in figure 5(c) the data of various typical Aurivillius ceramics [2, 3, 5, 13, 21, 24–30]. A sample with a higher d_{33} usually has a lower T_c, and it is evident that the CBT-*x*LiCe sample here has a rather large d_{33}.

In addition to effective poling and enhanced dipole stability, the reason why the highest d_{33} in CBT is achieved at a relatively low T_s may also be explained by its lattice structure (see figure 6). CBT has a tetragonal (space group *I*4/*mmm*) and an orthogonal (space group $A2_1am$) symmetry above and below T_c, respectively. The spontaneous polarization in BLSF ceramics mainly comes from the average shift of the Bi_mO_{3m+1} part of the perovskite slabs along the '*a*' axis and the displacement of the perovskite B-site cation relative to the surrounding oxygen octahedron, and these are associated with tilting, rotation and distortion of the oxygen octahedron due

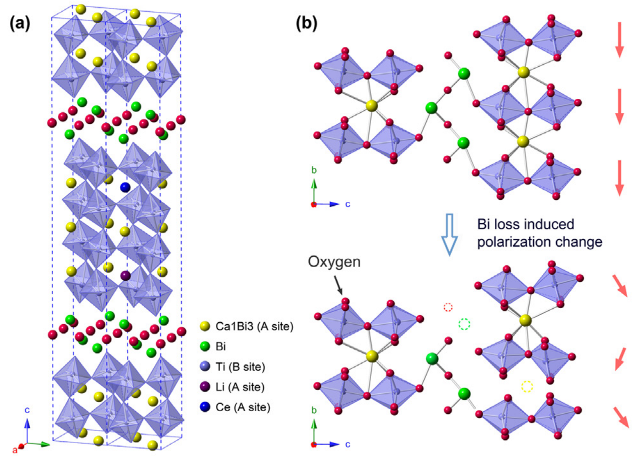

Figure 6. (a) Schematic illustration of the lattice structure of CBT-*x*LiCe; (b) illustration of influence of Bi deficiency on the dipole alignment, where the red arrows indicate dipoles within the *b*, *c*-plane.

Table 3. Composition of two typical samples determined by semi-quantitative analysis via XRF (the Li and Zn content are based on the raw material ratio).

Sintering conditions	Composition
920 °C, 4 h	$Ca_{0.95}Li_{0.027}Ce_{0.023}Bi_{3.98}Ti_{3.98}Zn_{0.01}O_{15}$
1080 °C, 4 h	$Ca_{0.94}Li_{0.023}Ce_{0.027}Bi_{3.46}Ti_{3.99}Zn_{0.01}O_{15}$

to strong bonding between Bi^{3+} and O^{2-} of the apex oxygen in the perovskite layer [4]. As a consequence, the polarization is largely restricted in the perovskite and is sensitive to lattice defects. Note that CBT consists of alternating $(Bi_2O_2)^{2+}$ and $(CaBi_2Ti_4O_{13})^{2-}$ layers, and two thirds of the perovskite A sites are occupied by Bi atoms. Thus, Bi deficiency could occur both in the $Bi_2O_2^{2+}$ layer and in the perovskite slab. We therefore propose that a slight Bi loss might significantly affect the spontaneous polarization due to the deformation and rotation of many connected octahedra in the perovskite slabs as schematically shown in figure 6. This assumption may well explain the change in P_r with increasing T_s and the critical role of Bi evaporation in controlling the piezoelectricity. To check the influence of T_s on Bi evaporation, the composition of two typical samples measured by XRF are shown in table 3, which indeed indicates that a substantial Bi deficiency appears at higher T_s.

4. Conclusions

An Aurivillius-type piezoelectric ceramic $(Ca_{1-2x}(LiCe)_xBi_4Ti_{3.99}Zn_{0.01}O_{15})$ with an ultrahigh Curie temperature (T_c) of about 800 °C and a significantly enhanced piezoelectric coefficient (d_{33}) comparable to those of textured ceramics was fabricated. Surprisingly, the highest d_{33} of 26 pC N^{-1} was achieved at a relatively low sintering temperature (T_s) of only 920 °C despite its density not being the highest. A relatively low T_s was found to be crucial for reducing Bi evaporation and suppressing abnormal grain growth, which are indispensable for high resistivity and effective poling due to decreased carrier density and restricted anisotropic conduction. Considering that the layered structure is sensitive to defects, we propose that controlled Bi loss is crucial for maintaining large polarization. This low-T_s system is very promising for industrial applications due to its enhanced piezoelectricity, low cost and high reproducibility. Contrary to the usual strategies of increasing density, this study reveals that a delicate balance of density, Bi loss and grain morphology by adjusting the sintering temperature is very effective for enhancing the performance of Aurivillius-type high-T_c ceramics.

Acknowledgments

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